

Module 5: Lesson 4

Bidirectional LSTMs and BTC prices



Outline

- ▶ The prediction problem at hand
- ▶ Bi-LSTM network for BTC returns

The prediction problem

Here, we repeat the same analysis we performed in the notebook of Lesson 2 of this module. In this case, however, we will deal with bidirectional LSTM (Bi-LSTM) instead of unidirectional ones. Our interest, one more time, lies in predicting short-term future asset returns:

- ▶ Which asset? **Bitcoin (BTC)**
- ▶ Which returns? **25-hour ahead returns**
- ▶ Which inputs? Past cumulative returns for **1-hour, 10-hour, and 50-hour**
- ▶ Which period? **Jan-2021 to Sept-2022**

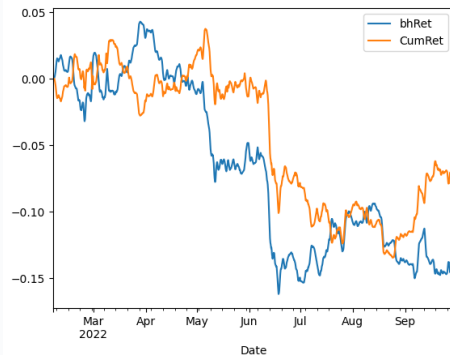
	date	Ret	Ret_10	Ret_50	Ret25
52634	2021-01-03 01:00:00	0.003812	0.000300	0.001236	0.000294
52635	2021-01-03 02:00:00	0.004669	0.000103	0.001244	0.000270
52636	2021-01-03 03:00:00	-0.004577	-0.000065	0.001066	0.000233
52637	2021-01-03 04:00:00	0.019479	0.000120	0.001334	-0.000088
52638	2021-01-03 05:00:00	0.014644	0.000235	0.001445	-0.000118

Bi-LSTM model to predict BTC returns

In the notebook accompanying this Lesson, we will guide you through the development of a Bi-LSTM model with bi. Here you have the summary of the model, as well as the out-of-sample performance of a backtest for the strategy using model predictions:

Layer (type)	Output Shape	Param #
bidirectional (BidirectionalL)	(None, 30, 100)	21600
bidirectional_1 (BidirectionalL)	(None, 30, 100)	60400
dropout (Dropout)	(None, 30, 100)	0
bidirectional_2 (BidirectionalL)	(None, 30, 100)	60400
dropout_1 (Dropout)	(None, 30, 100)	0
bidirectional_3 (BidirectionalL)	(None, 30, 100)	60400
dropout_2 (Dropout)	(None, 30, 100)	0
bidirectional_4 (BidirectionalL)	(None, 100)	60400
dropout_3 (Dropout)	(None, 100)	0
dense (Dense)	(None, 20)	2020
dropout_4 (Dropout)	(None, 20)	0
dense_1 (Dense)	(None, 10)	210
dropout_5 (Dropout)	(None, 10)	0
dense_2 (Dense)	(None, 1)	11

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Total params: 265,441
Trainable params: 265,441
Non-trainable params: 0



Summary of Lesson 4

In Lesson 4, we have:

- ▶ Developed our first bidirectional LSTM Neural Network model
- ▶ Compared the complexity of bidirectional models with unidirectional ones
- ▶ Evaluated the predictive power of bidirectional LSTM and backtest the strategy in test sample.

TO DO NEXT: Go to the accompanying Python Notebook to implement your first bidirectional LSTM!

In the next module, we will introduce Autoencoders, an example of unsupervised neural network.